

Deep Generative Models

Lecture 13

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AI Masters

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Recap of Previous Lecture

Let us choose $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$. Then $p(\mathbf{z}) = p(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$.

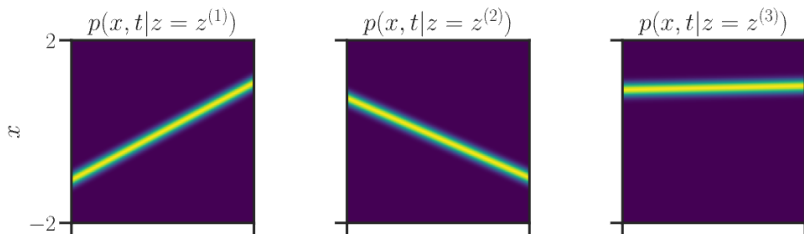
$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Let's consider straight conditional paths:

$$\boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \quad \boldsymbol{\sigma}_t(\mathbf{x}_1) = \epsilon$$



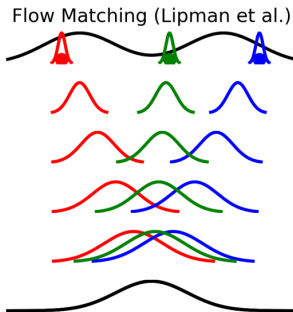
Recap of Previous Lecture

Endpoint conditioning

$$\mathbf{z} = \mathbf{x}_1$$

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

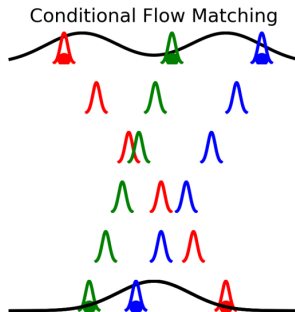


Pair conditioning

$$\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$$

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$



Recap of Previous Lecture

- ▶ This conditioning allows us to transport any distribution $p_0(\mathbf{x})$ to any distribution $p_1(\mathbf{x})$.
- ▶ It's possible to apply this approach to paired tasks, e.g., style transfer.

Training Procedure

1. Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$.
2. Sample time $t \sim U[0, 1]$.
3. Compute the noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
4. Compute the loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim p_0(\mathbf{x})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1)$$

Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

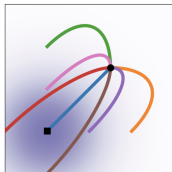
$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

Variance Exploding SDE

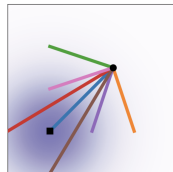
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\mathbf{x}_1, \sigma_{1-t}^2 \mathbf{I}) \quad \Rightarrow \quad \mathbf{v}(\mathbf{x}, \mathbf{x}_1, t) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}(\mathbf{x}_t - \mathbf{x}_1)$$

Variance Preserving SDE

$$p_t(\mathbf{x}_t|\mathbf{x}_1) = \mathcal{N}(\alpha_{1-t}\mathbf{x}_1, (1 - \alpha_{1-t}^2)\mathbf{I}) \quad \Rightarrow \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\alpha'_{1-t}}{1 - \alpha_{1-t}^2} \cdot (\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$



Diffusion



OT

Recap of Previous Lecture

Continuous state space

- ▶ **Discrete time** $t \in \{0, 1, \dots, T\} \Rightarrow$ **DDPM / NCSN.**
- ▶ **Continuous time** $t \in [0, 1] \Rightarrow$ **Score-based SDE models.**

Discrete state space

- ▶ **Discrete time** $t \in \{0, 1, \dots, T\}.$
- ▶ **Continuous time** $t \in [0, 1].$

Key advantages of discrete diffusion

- ▶ Parallel generation
- ▶ Flexible infilling
- ▶ Robustness
- ▶ Unified framework

Recap of Previous Lecture

Discrete Diffusion Markov Chain

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \text{Cat}(\mathbf{Q}_t \mathbf{x}_{t-1}),$$

Each $\mathbf{x}_t \in \{0, 1\}^K$ is a **one-hot vector** encoding the categorical state (it is just one token).

Transition Matrix

$$[\mathbf{Q}_t]_{ij} = q(x_t = i | x_{t-1} = j), \quad \sum_{i=1}^K [\mathbf{Q}_t]_{ij} = 1.$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{x}_0), \quad \mathbf{Q}_{1:t} = \mathbf{Q}_t \mathbf{Q}_{t-1} \cdots \mathbf{Q}_1.$$

- ▶ The choice of $\mathbf{Q}_t$ determines how information is erased and what the stationary distribution becomes.
- ▶ $\mathbf{Q}_t$ and $\mathbf{Q}_{1:t}$ should be easy to compute for each t .

Recap of Previous Lecture

Uniform vs. Absorbing Transition Matrix

Aspect	Uniform Diffusion	Absorbing Diffusion
$\mathbf{Q}_t$	$(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$	$(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:t}$	$\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{U}$	$\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:\infty}$	$\mathbf{U}$	$\text{Cat}(\mathbf{e}_m)$
Interpretation	Random replacement	Gradual masking of tokens
Application	Image diffusion	Text diffusion $\approx$ Masked LM

Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

Outline

1. Discrete Diffusion

- Reverse Discrete Diffusion

- From Token to Sequence

- Absorbing Diffusion

- Continuous-Time Formulation

2. Course Overview

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Posterior of the Forward Process

ELBO

$$\begin{aligned}\mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

Posterior of the Forward Process

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- ▶ Conditioned reverse distribution $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ played crucial role in the continuous-state diffusion model.
- ▶ It shows the probability of a previous state given the noisy state $\mathbf{x}_t$ and the original clean data $\mathbf{x}_0$.

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Discrete conditioned reverse distribution

$$\begin{aligned} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\ &= \frac{\text{Cat}(\mathbf{Q}_t) \cdot \text{Cat}(\mathbf{Q}_{1:t-1})}{\text{Cat}(\mathbf{Q}_{1:t})}. \end{aligned}$$

Posterior of the Forward Process

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$

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Recall the ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}} \left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \right),$$

Posterior of the Forward Process

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Recall the ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)),$$

- ▶ Both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $q(\mathbf{x}_t|\mathbf{x}_0)$ are known analytically from the forward process.
- ▶ The reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a learned categorical distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}(\boldsymbol{\pi}_\theta(\mathbf{x}_t, t)),$$

where $\boldsymbol{\pi}_\theta$ is a neural network.

Discrete-time ELBO for Discrete Diffusion

ELBO term

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Categorical KL

$$D_{\text{KL}}(\text{Cat}(\mathbf{q}) \parallel \text{Cat}(\mathbf{p})) = \sum_{k=1}^K q_k \log \frac{q_k}{p_k} = H(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}),$$

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- ▶ $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$ is a constant w.r.t. θ .
- ▶ $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$ is a **cross-entropy loss**.

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- ▶ $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$ is a constant w.r.t. θ .
- ▶ $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$ is a **cross-entropy loss**.

Therefore, minimizing $\mathcal{L}_t$ w.r.t. θ is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0), p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Outline

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Reverse Discrete Diffusion

From Token to Sequence

Absorbing Diffusion

Continuous-Time Formulation

2. Course Overview

From Token to Sequence

One-hot sequence representation

$$\mathbf{x}_t \in \{0, 1\}^K \quad \Leftrightarrow \quad \mathbf{X}_t \in \{0, 1\}^{K \times m}$$

Here $\mathbf{X}_t$ is a one-hot representation of a sequence of tokens.

From Token to Sequence

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Independent Token-wise Forward Process

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

- ▶ Each position i evolves according to its own Markov chain.
- ▶ Often the same transition matrix $\mathbf{Q}_t$ is shared across i .

From Token to Sequence

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Continuous Diffusion Analogy

- ▶ In Gaussian DDPMs with diagonal covariance, noise is independent per pixel.
- ▶ Structure is not in the noise; it is learned by the reverse model.

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Conditioned Reverse Distribution

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$

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- ▶ All distributions defined by the forward process are factorized.
- ▶ Dependence appears in the learned reverse model.

Reverse Model for Sequence

$$p_{\theta}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

- ▶ The output factorizes (parallel prediction across positions).
- ▶ Each factor conditions on the entire noisy sequence $\mathbf{X}_t$.
- ▶ This is exactly the **masked language modeling** pattern.

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Objective: $\mathcal{L}_t$ term

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Final objective: masked LM

$$\mathcal{L} = \sum_{t=1}^T \sum_{i=1}^m \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[-\log p_{\theta}(\mathbf{x}_0^i|\mathbf{X}_t) \right].$$

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Absorbing Diffusion: Forward Process

Let's restrict to the case of absorbing transition matrix.

$$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top, \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s).$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t \mathbf{I} + (1 - \bar{\alpha}_t) \mathbf{e}_m \mathbf{1}^\top.$$

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Each position is either still clean or already masked:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \bar{\alpha}_t [\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t) [\mathbf{x}_t = \mathbf{e}_m]$$

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$$\mathbf{Q}_t = \begin{pmatrix} 1 - \beta_t & 0 & 0 \\ 0 & 1 - \beta_t & 0 \\ \beta_t & \beta_t & 1 \end{pmatrix} \Rightarrow \text{the masked state is absorbing.}$$

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What happens in the conditioned reverse process $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$?

Absorbing Diffusion

Conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} [\mathbf{x}_{t-1} = \mathbf{x}_t], & \text{if } \mathbf{x}_t \neq \mathbf{e}_m, \\ \rho_t[\mathbf{x}_{t-1} = \mathbf{x}_0] + (1 - \rho_t)[\mathbf{x}_{t-1} = \mathbf{e}_m], & \text{if } \mathbf{x}_t = \mathbf{e}_m, \end{cases}$$

where

$$\rho_t = \frac{\beta_t \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}.$$

Absorbing Diffusion

Conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} [\mathbf{x}_{t-1} = \mathbf{x}_t], & \text{if } \mathbf{x}_t \neq \mathbf{e}_m, \\ \rho_t[\mathbf{x}_{t-1} = \mathbf{x}_0] + (1 - \rho_t)[\mathbf{x}_{t-1} = \mathbf{e}_m], & \text{if } \mathbf{x}_t = \mathbf{e}_m, \end{cases}$$

where

$$\rho_t = \frac{\beta_t \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}.$$

- ▶ If $\mathbf{x}_t \neq \mathbf{e}_m$, then the token must be unchanged: $\mathbf{x}_{t-1} = \mathbf{x}_t$.
- ▶ Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \dots = \mathbf{x}_0$.

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- ▶ Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \dots = \mathbf{x}_0$.
- ▶ If $\mathbf{x}_t = \mathbf{e}_m$, the previous token may be either clean or masked.
- ▶ With probability ρ_t , masking occurred exactly at step t (so $\mathbf{x}_{t-1} = \mathbf{x}_0$).
- ▶ With probability $(1 - \rho_t)$, the token was already masked earlier (so $\mathbf{x}_{t-1} = \mathbf{e}_m$).

Absorbing Diffusion

Sequence Distribution

$$q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^m q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i).$$

Each position i has two possible cases:

$$\mathbf{x}_t^i \neq \mathbf{e}_m \Rightarrow \mathbf{x}_{t-1}^i = \mathbf{x}_t^i, \quad \mathbf{x}_t^i = \mathbf{e}_m \Rightarrow \mathbf{x}_{t-1}^i \in \{\mathbf{x}_0^i, \mathbf{e}_m\}.$$

Absorbing Diffusion

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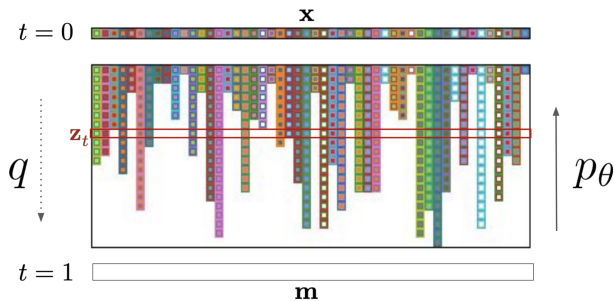
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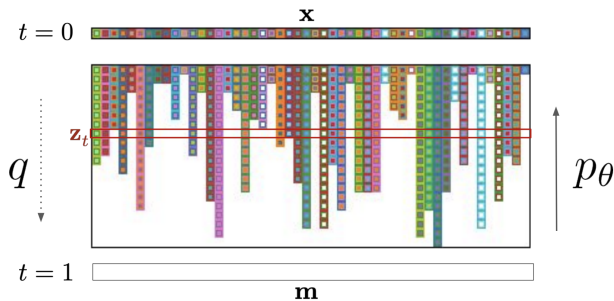
Interpretation

- ▶ The forward process produces **random partial observations** of the clean sequence.
- ▶ If a token is visible at time t , the reverse distribution is deterministic.
- ▶ Unmasked tokens yield a deterministic posterior and therefore contribute only a constant to the ELBO. Therefore, only masked tokens contribute to the training loss.

Absorbing Diffusion

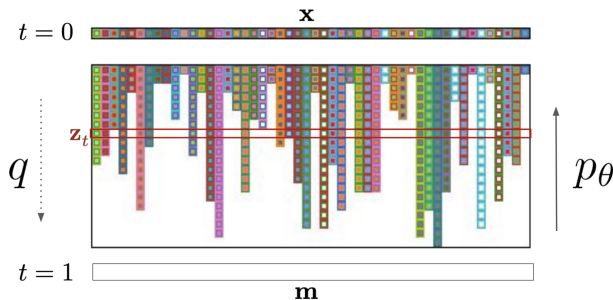


Absorbing Diffusion



$$p_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^m p_\theta(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

Absorbing Diffusion



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Objective: sequence-level $\mathcal{L}_t$

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^m \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_t) \right] + \text{const}$$

Algorithm: Discrete-Time Absorbing Diffusion

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$ and $t \sim U\{1, \dots, T\}$.
2. Corrupt the sequence by independent masking:

$$\mathbf{x}_t^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } 1 - \bar{\alpha}_t, \\ \mathbf{x}_0^i, & \text{with prob. } \bar{\alpha}_t, \end{cases} \quad i = 1, \dots, m.$$

3. Predict token distributions in parallel:

$$p_{\theta}(\mathbf{X}_0 | \mathbf{X}_t) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_0^i | \mathbf{X}_t).$$

4. Compute the masked-CE loss:

$$\mathcal{L}(\theta) = \rho_t \sum_{i=1}^m [\mathbf{x}_t^i = \mathbf{e}_m] [-\log p_{\theta}(\mathbf{x}_0^i | \mathbf{X}_t)].$$

Algorithm: Discrete-Time Absorbing Diffusion

Sampling

1. Initialize $\mathbf{X}_T \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).

2. For $t = T, T - 1, \dots, 1$:

2.1 Predict $p_\theta(\mathbf{x}_0^i | \mathbf{X}_t)$ for all positions.

2.2 For each masked position ($\mathbf{x}_t^i = \mathbf{e}_m$):

$$\mathbf{x}_{t-1}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_\theta(\mathbf{x}_0^i | \mathbf{X}_t), & \text{with prob. } \rho_t, \\ \mathbf{e}_m, & \text{with prob. } 1 - \rho_t. \end{cases}$$

2.3 For each unmasked position: $\mathbf{x}_{t-1}^i = \mathbf{x}_t^i$.

3. Return the final sequence $\mathbf{X}_0$ (fully unmasked).

Outline

1. Discrete Diffusion

Reverse Discrete Diffusion

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2. Course Overview

From Discrete Time to Mask Rate

In absorbing diffusion, the forward process is

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m].$$

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- The distribution depends on t only through the scalar

$$\lambda_t = 1 - \bar{\alpha}_t \in [0, 1].$$

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- ▶ We can therefore reparameterize the corruption level by

$$t \quad \Rightarrow \quad \lambda \in [0, 1].$$

- ▶ We directly define a family of corrupted distributions indexed by a continuous mask rate λ :

$$q(\mathbf{x}_\lambda|\mathbf{x}_0) = (1 - \lambda)[\mathbf{x}_\lambda = \mathbf{x}_0] + \lambda[\mathbf{x}_\lambda = \mathbf{e}_m].$$

Discrete ELBO Revisited

Recall the per-step ELBO term for absorbing diffusion:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^m \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_{\theta}(\mathbf{x}_0^i|\mathbf{X}_t) \right] + \text{const.}$$

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Replacing the discrete index t with the continuous mask rate λ , the training objective becomes

$$\mathcal{L} = \int_0^1 w(\lambda) \mathbb{E}_{q_\lambda(\mathbf{x}_\lambda|\mathbf{x}_0)} \sum_{i=1}^m [\mathbf{x}_\lambda^i = \mathbf{e}_m] [-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_\lambda)] d\lambda.$$

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Interpretation

Training corresponds to optimizing a **continuous mixture of masked language modeling objectives** with different mask rates.

Algorithm: Masked Diffusion Language Model (MDLM)

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$ and $\lambda \sim U[0, 1]$.
2. Corrupt the sequence by independent masking:

$$\mathbf{x}_\lambda^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } \lambda, \\ \mathbf{x}_0^i, & \text{with prob. } 1 - \lambda, \end{cases} \quad i = 1, \dots, m.$$

3. Predict token distributions in parallel:

$$p_\theta(\mathbf{X}_0 | \mathbf{X}_\lambda) = \prod_{i=1}^m p_\theta(\mathbf{x}_0^i | \mathbf{x}_\lambda^i).$$

4. Compute the masked-CE loss:

$$\mathcal{L}(\theta) = w(\lambda) \sum_{i=1}^m [\mathbf{x}_\lambda^i = \mathbf{e}_m] [-\log p_\theta(\mathbf{x}_0^i | \mathbf{x}_\lambda^i)].$$

Algorithm: Masked Diffusion Language Model (MDLM)

Sampling

1. Initialize $\mathbf{X}_{\lambda_1} \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).
2. For a decreasing schedule $1 = \lambda_1 > \lambda_2 > \dots > \lambda_L = 0$, $\ell = 1, \dots, L - 1$:
 - 2.1 Predict $p_\theta(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell})$ for all positions.
 - 2.2 For each masked position ($\mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m$):

$$\mathbf{x}_{\lambda_{\ell+1}}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_\theta(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell}), & \text{with prob. } 1 - \frac{\lambda_{\ell+1}}{\lambda_\ell}, \\ \mathbf{e}_m, & \text{with prob. } \frac{\lambda_{\ell+1}}{\lambda_\ell}. \end{cases}$$

- 2.3 For each unmasked position: $\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{x}_{\lambda_\ell}^i$.
3. Return the final sequence $\mathbf{X}_{\lambda_L}$ (fully unmasked).

Note: Once unmasked, a token stays unmasked. So for a currently masked token:

$$\mathbb{P}(\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{e}_m \mid \mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m) = \frac{\lambda_{\ell+1}}{\lambda_\ell}.$$

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Course Overview: Problem Statement

Goal

Learn a generative model $p_{\theta}(\mathbf{x})$ that matches the data distribution $p_{\text{data}}(\mathbf{x})$.

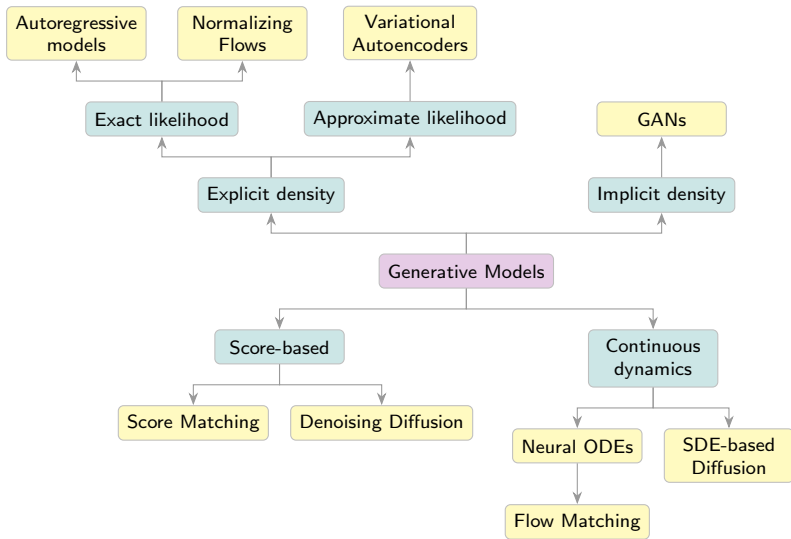
Three lenses on the same problem

- **Divergence minimization** (L1, L5):

$$\min_{\theta} D(p_{\text{data}} \parallel p_{\theta}) \quad (\text{KL, JS, Wasserstein, } \dots)$$

- **Likelihood-based** (L1–L4, L7–L8): maximize $\log p_{\theta}(\mathbf{x})$ (AR, NF, VAE, diffusion as hierarchical VAE).
- **Score-based** (L6–L10): learn $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ (SM / NCSN, DDPM, score-based SDE).

Course Roadmap



Likelihood-based: Exact & Approximate Density (L1–L4)

Autoregressive (L1)

$$p_{\theta}(\mathbf{x}) = \prod_{i=1}^m p_{\theta}(x_i | \mathbf{x}_{1:i-1}) \quad \text{exact likelihood, sequential sampling}$$

Normalizing Flows (L2)

$$\mathbf{x} = \mathbf{f}_{\theta}(\mathbf{z}), \quad \log p_{\theta}(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$$

VAE / VQ-VAE (L3–L4)

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$$

- ▶ Amortized inference + reparametrization trick.
- ▶ ELBO surgery $\Rightarrow$ optimal prior; VQ-VAE with discrete latents.

Implicit & Score-based Foundations (L5–L6)

GAN / WGAN (L5)

- ▶ Implicit p_θ , adversarial min-max

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))$$

- ▶ Optimal discriminator $\Rightarrow$ JSD; WGAN uses Wasserstein distance via Kantorovich–Rubinstein duality.
- ▶ Evaluation tools: FID, Precision/Recall, CLIP score, human eval.

Score Matching / NCSN (L6)

- ▶ Learn $\mathbf{s}_{\theta}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ via denoising SM.
- ▶ Sample with (annealed) Langevin dynamics over multiple noise scales.

Diffusion Models (L7–L8)

Gaussian diffusion (L7)

- ▶ Forward $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, direct sampling $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$.
- ▶ Diffusion as hierarchical VAE; ELBO splits into reconstruction + prior KL + $\sum_t \mathcal{L}_t$.

DDPM (L8)

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left\| \epsilon - \epsilon_{\theta}(\mathbf{x}_t, t) \right\|^2, \quad \epsilon_{\theta} \propto -\sigma_t \mathbf{s}_{\theta}$$

- ▶ Equivalent to score-based generative modeling (NCSN-style).

Guidance (L8)

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t, t, \mathbf{y}) = (1 - \gamma) \epsilon_{\theta}(\mathbf{x}_t, t, \emptyset) + \gamma \epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{y})$$

Continuous Dynamics & SDE (L9–L10)

Neural ODE / continuous-time NF (L9)

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t), \quad \frac{d}{dt} \log p_t(\mathbf{x}(t)) = -\text{tr}\left(\frac{\partial \mathbf{v}_{\theta}}{\partial \mathbf{x}}\right)$$

Forward SDE (L9)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

- KFP equation; VE / VP SDEs unify NCSN and DDPM.

Reverse SDE & Probability flow ODE (L10)

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t) \mathbf{s}_{\theta}(\mathbf{x}, t)] dt + g(t) d\bar{\mathbf{w}}$$
$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t) \mathbf{s}_{\theta}(\mathbf{x}, t) \quad (\text{deterministic equivalent})$$

Flow Matching (L10–L11)

Flow Matching objective (L10)

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, p_t(\mathbf{x})} \left\| \mathbf{v}_{\theta}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, t) \right\|^2$$

Target $\mathbf{v}(\mathbf{x}, t)$ is the marginal vector field — intractable in general.

Conditional Flow Matching (L11)

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, p(\mathbf{z}), p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}_{\theta}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \right\|^2$$

- ▶ $\nabla_{\theta} \mathcal{L}_{\text{FM}} = \nabla_{\theta} \mathcal{L}_{\text{CFM}}$ — regress against *conditional* field.
- ▶ Conditioning on endpoint $\mathbf{z} = \mathbf{x}_1$ (one-sided) or pair $(\mathbf{x}_0, \mathbf{x}_1)$ (two-sided).
- ▶ Straight Gaussian paths $\Rightarrow$ closed-form $\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \mathbf{x}_1 - \mathbf{x}_0$.

Discrete Diffusion (L12–L13)

Forward discrete process (L12)

Categorical $\mathbf{x}_t$ with transition matrices $\mathbf{Q}_t$:

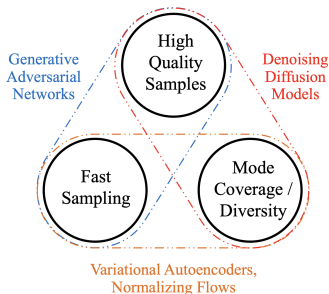
- ▶ **Uniform:** $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$
- ▶ **Absorbing:** $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$ (mask token)

Reverse process & Discrete ELBO (L13)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t\mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right),$$

each $\mathcal{L}_t$ reduces to a **cross-entropy** loss against $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$.

Generative Learning Trilemma



Rule of thumb

- ▶ **Likelihood + Coverage** $\Rightarrow$ **AR / NF**
exact density, *slow sampling*.
- ▶ **Likelihood + Fast Sampling** $\Rightarrow$ **VAE**
tractable bound, *blurry samples*.
- ▶ **Quality + Fast Sampling** $\Rightarrow$ **GAN**
sharp, *mode collapse, no likelihood*.
- ▶ **Quality + Coverage** $\Rightarrow$ **Diffusion**
stable training, *slow sampling*.
- ▶ **Quality + Faster Sampling** $\Rightarrow$ **FM / ODE**
fewer steps, *approx. likelihood*.
- ▶ **Discrete + Coverage** $\Rightarrow$ **Discrete diffusion**
parallel denoising, *iterative decoding*.

Summary

- ▶ Reverse discrete diffusion process uses the variational approach to invert forward process; the discrete-state ELBO is a cross-entropy loss.
- ▶ In the absorbing case only masked tokens contribute to the ELBO, which reduces to a continuous mixture of MLM losses over the mask rate $\lambda \in [0, 1]$.
- ▶ MDLM sampling performs iterative parallel refinement from fully masked to fully unmasked sequences.
- ▶ No generative model dominates all axes of the trilemma — each family trades off likelihood, sample quality, mode coverage and sampling speed.
- ▶ Modern frontier (flow matching, discrete diffusion, distilled diffusion) tries to push the Pareto frontier of this trilemma.